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(1) Research Program/Project/Study Title: Body Shape Analysis of Bayuyan Shells using Geometric Morphometric Analysis in Capiz, Philippines	
Proponent Name:	STEPHANIE S. PIMENTEL, PhD
Academic Rank:	
Co-Authors (with academic rank):	
Campus/Satellite College/Address:	FFRDC
Telephone/Cell Phone/Fax Number:	
E-mail Address:	sspimentel@capsu.edu.ph

Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
		Municipality/City:	
		Province:	
		Region:	
(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☐ Applied ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☐ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
(7) Sector/Commodity		(8) Discipline/Field of Specialization	

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(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

Molluscs with bivalve morphology are renowned for their extraordinary diversity. Their sturdy, rigid, and stable shells, which serve as both support and protection, define them (Leyva-Valencia et al., 2012). Research on their morphology shells and morphometric relationships has been applied to various tasks, such as monitoring stocks, analyzing sexual dimorphism, and prosize-viding-based assessments of communities (Robinson et al., 2010). Since the shape of bivalve species is known to change depending on their growing environment, establishing a link between shape and location can be helpful for various ecologically pertinent issues, such as traceability (Pepin & Carr, 1993; Swain et al., 2001; Higgins et al., 2010), conservation, and monitoring (Rufino et al., 2012; Morais et al., 2014; Aguirre et al., 2015). Numerous external (biotic and abiotic interactions) and endogenous (genetic and physiological) factors influence the morphology of bivalve shells (Scholz & Hartman, 2007; Van Bocxlaer & Van Damme, 2009; Zieritz & Aldridge, 2009). Determining the connections between the morphology of bivalve species and the features of their habitats is essential to conservation efforts. This is because moving specimens for repopulation or altering the environment might severely compromise a bivalve's chances of survival (Hornbach et al., 2010).

The study of trait variation and its covariance with other variables is known as morphometrics (Rohlf and Slice, 1990). Nowadays, biological classification (Minton et al. 2008, Perez 2011, Miller 2016), botany (Su et al. 2021), and medicine (Du and Lu 2006) all make significant use of geometric morphometric (GM) approaches. Using genetic modification, Shu et al. (2022) investigated the morphological variations among eight Chinese scallop species. GM was successfully employed by Jiang et al. (2019) to distinguish between various regional populations of Chinese mitten crabs (*Eriocheir sinensis*).

Bayuyan shells biology

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The general objective of this study is to determine the body morphometry of Bayuyan Shells using Geometric Morphometric Analysis in Panay, Capiz, Philippines. Specifically, it aims to:

1. Determine the body shape of Bayuyan shells using Geometric Morphometric Analysis collected from Insaygan and Butacal, Panay, Capiz;
2. Delineate the body shape of the Bayuyan Shells collected from the two aforementioned sites;
3. Identify the specific body shape of Bayuyan shells through pre-determined landmarks.

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(11) Materials and Methods or Methodology

Study Site

The study site was located at the coastal barangays of Panay, Capiz (Fig. 1) - the Seafood Capital of the Philippines - in the Northern part of Panay Island. The barangays stated were the province's major shellfish source and other marine resources. Samples were collected by the locals of the two aforementioned sites. Thirty (30) samples were collected from Insaygan and thirty samples were also collected from Butacal, Panay, Capiz.

Sample Collection

Sixty samples were collected from the two sampling locations from March 2024 to June 2024. Upon sampling collection, samples were transported to the Fisheries and Food Research Development Center Laboratory to be cleaned. Meristic characteristics were gathered. Samples were then carefully opened to examine and determine the meat weight. The shells were then sundried for 24 hours. Cleaned samples were then subjected to photographed individually using a 108MP phone camera and compiled in a file to be processed in Thinplate Spline. Shell samples were put in sealed plastic bags and labeled to be kept.

Landmark-based Geometric Morphometrics

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Sixteen pre-determined landmark points were used to determine the body morphometry of Bayuyan shells collected from the two sites - Insaygan and Butacal located at the Municipality of Panay. Figure 2 depicts the locations of the landmark points and its descriptive location (as shown in Table 1).

Data Analysis

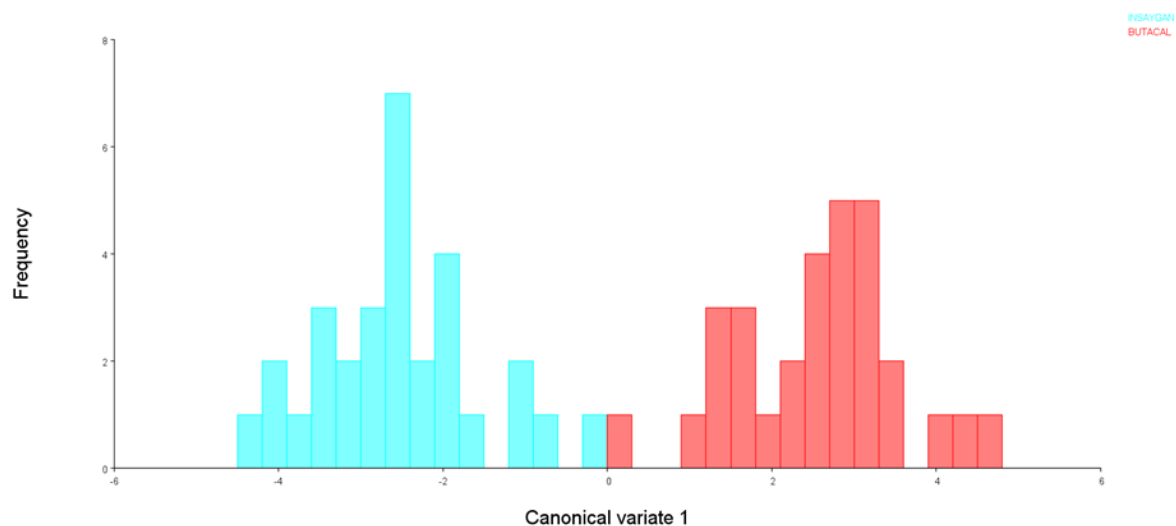
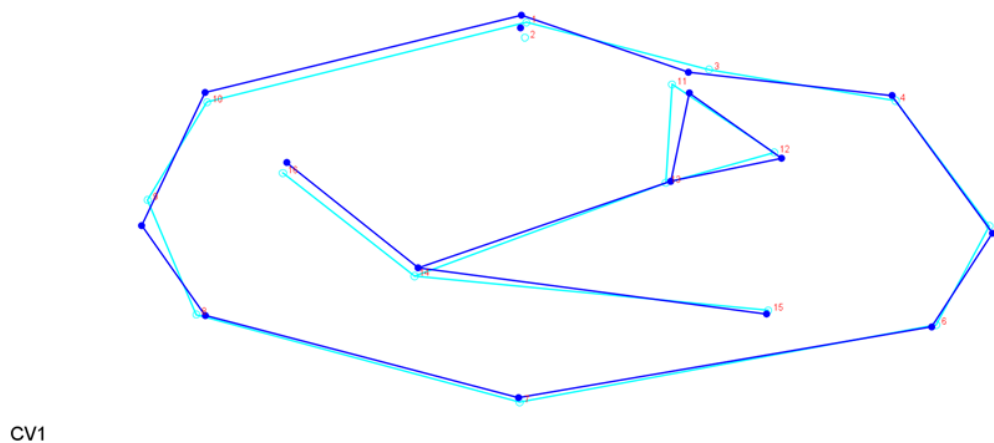
Thirty (30) samples were collected from the two sampling sites in Capiz, Philippines. Each of the samples were photographed. Photos obtained were compiled and subjected to data processing. Compiled photos were saved as a TPS file using a thinplate spline application (tps) application which controls and organizes the display, functioning, and storage of TPS files by applying the random scrambling scenarios to minimize bias in digitizing landmark locations, restoring the original order for repeated statistical analysis, creating empty files and adding keywords, etc. TPSUtil was used to convert the photos into TPS files. The files were subjected to digitalization and landmarks through the tpsDig application. TPSDig is a Windows (rough 11) program for digitizing landmarks and outlines for geometric morphometric analyses. Includes simple image enhancement operations, scale factors, image brightness profile, and support for AVI and MOV files. Output is in the TPS file format used by the "tps series" of programs. After digitalization, the files were then subjected to data analysis through MorphoJ. This application provides a user-friendly platform for the most important types of analyses in geometric morphometrics. The program is written for 2D and 3D data, with inevitable differences in the graphical output possibilities.

(12) Results and Discussions

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Body Shape of Bayuyan Shells Collected from Insaygan and Butacal, Panay, Capiz

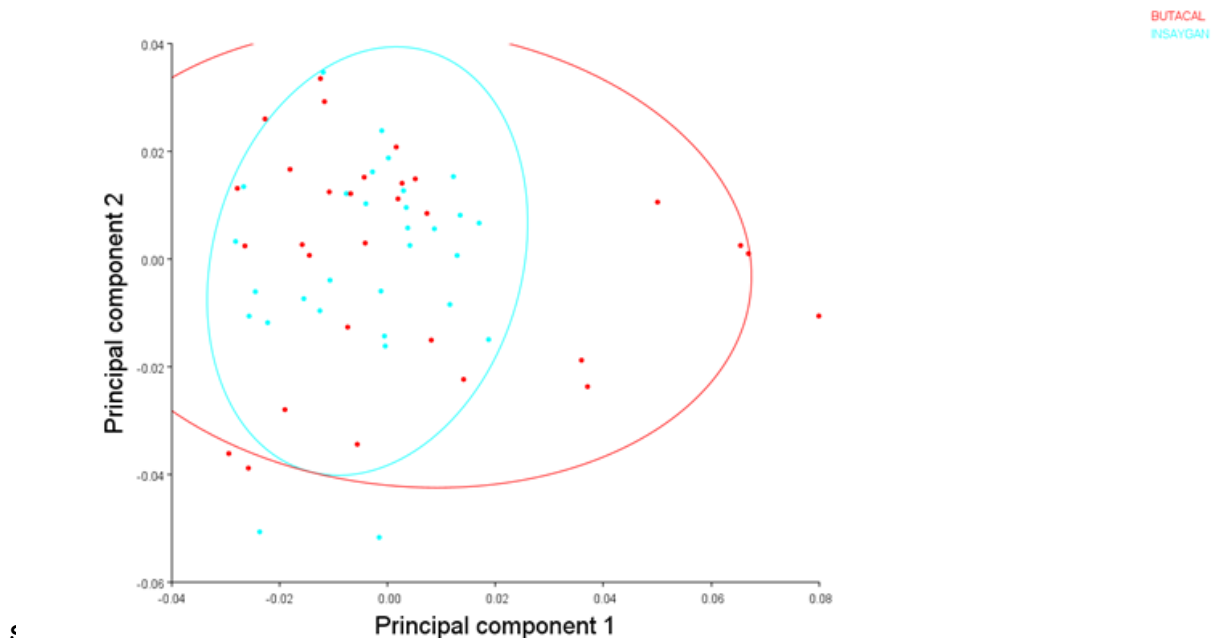
Sixty samples of Bayuyan shells were subjected to digitalization using tps applications developed by Rolph. The results of Canonical Variance Analysis (Fig 2) show a slight variation among the population of Bayuyan shells in terms of body morphometry. The variation was observed at landmarks 8 () and 9 (). According to Woods (1999), these irregular patterns that are observed in most burrowing bivalves are due to their feeding behavior and escape mechanism of the organisms. They burrow themselves in the sediments, causing friction between the tip of the shell and the sediment.



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Shape Variation Among the Population of Bayuyan Shells in terms of Location

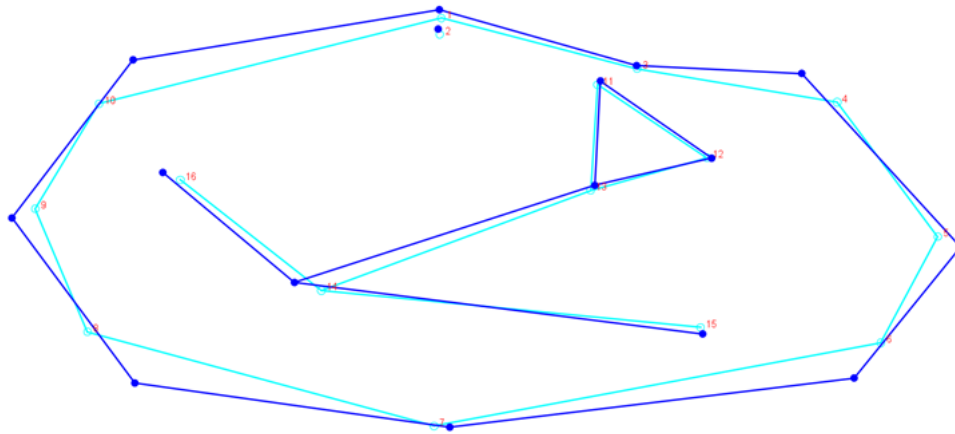
Results of Principal Component Analysis determined the relationship between the samples collected from Butacal and Insaygan, Panay, Capiz. The result (Figure 3) depicts an overlapping of the population of Bayuyan shells. This indicates that the samples were from the same species (Rohlf, 1999). This claim was supported by Mahalanobis Distance of 5.1113 with a p-value of <.0001 and Procrustes Distance of 0.0219 with a p-value of <.0001.



Principal Component Analysis determined the specific body shape of Bayuyan shells collected from the two aforementioned sites - Butacal and Insaygan, Panay, Capiz. The PCA results show a variation in the landmarks located at the tip anterior and posterior region of the shells of Bayuyan. Landmark 4 (), Landmark 5 (), Landmark 6 (), Landmark 8 (), Landmark (9), and Landmark 10 () showed the most variation. These regions are the ones that are responsible for the burrowing and procurement of food by most of the bivalves (Salvacion et al., 2017). As stated by Aranov et al. (2010), based on the mechanism of food collection, bivalves can be suspension-feeders or deposit-feeders or even utilize both feeding methods. Bivalves also burrow in the sediment to protect themselves from the predators.

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PC1



(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indentation with 12-point space in between.)

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